

A mechanism for the deformation of disordered states of matter

K.A. Padmanabhan^{a,*}, H. Gleiter^b

^a*School of Engineering Sciences & Technology and Centre for Nanotechnology, University of Hyderabad, Hyderabad 500 046, India*

^b*KIT Campus North, Institute of Nanotechnology, P.O. Box 3640, 0721 Karlsruhe, Germany*

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ABSTRACT

As the frontier in advanced materials development has shifted into highly disordered systems, concepts of deformation based on crystal lattice dislocations often become too coarse to be of relevance. Therefore, a new deformation process, localized to dimensions smaller than those involved in dislocation mechanisms, was proposed sometime ago. Some of its important features are discussed here to suggest that this mechanism is likely to be of use in understanding the superplastic deformation of metals and alloys, ceramics, metal-matrix- and ceramic-matrix-composites, dispersion hardened materials, intermetallics, geological materials, metallic glasses and poly-glasses of grain sizes in the μm -, sub- μm - or nm-range – a much wider area of application than originally anticipated. This will allow one to define “superplasticity” as due to a unique physical mechanism, rather than by the extreme elongations obtainable in tensile testing or the strain rate sensitivity index being more than ~ 0.30 .

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1. Introduction

Advanced materials development has moved on from slightly imperfect materials to highly disordered systems, e.g., glassy alloys, vapor-quenched or rapidly solidified materials, sputtered films, superplastic alloys, micro-, sub-micro- and nano-crystalline materials. “The concept of a crystal dislocation loses much of its value once there is no longer a good long-range crystal structure in which to define it. For breach of order, there must first be a framework of order” [1].

The phenomenon of superplasticity, during the occurrence of which specimens subjected to a small tensile stress can be stretched by several hundred percent, has emerged as a near-ubiquitous phenomenon, present within well-defined grain size – strain rate – temperature domains in metals and alloys, ceramics, metal-matrix- and ceramic-matrix-composites, dispersion hardened materials, intermetallics, geological materials and metallic glasses. The phenomenon is also present in microcrystalline, sub-microcrystalline and nanocrystalline forms of these classes of materials. There is near-universal agreement that bulk of the superplastic deformation is traceable to grain boundary/interphase interface sliding [2–7].

The aim of this paper is to highlight some of the features of a mechanism proposed earlier [8–20] to suggest that that mechanism could be important in all situations where concepts based on dislocations are of limited use. Therefore, an attempt will be

made to explain the phenomenon of superplasticity in terms of a single mechanism of deformation, which applies equally well to the different kinds of disordered states of matter listed above. Needless to say, under unfavorable experimental conditions, less strain-rate sensitive mechanisms would intervene and superplastic elongations may not result even in materials that are known to exhibit superplasticity.

In the present viewpoint, the atomistic details of a unit boundary sliding event are considered to be the same regardless of whether one is concerned with a material of grain size in the micrometer, sub-micrometer or nanometer range in the different classes of materials. The distribution/arrangement of atoms and free volume fraction within the basic unit of sliding, the presence of free volume sites along the boundary and the propagation of deformation along the interfaces in a given material, however, will depend on the atomic configurations present in the basic unit of sliding (which are governed, among other things, by the inter-atomic forces) and its neighborhood in the boundary. To advocate the strength in our arguments, we use the mathematical proof provided by Hutter [21] that “shorter computable theories have more weight when calculating the expected value of an action across all computable theories which perfectly describe previous observations”. The applicability of the physics underlying the model to all the classes of materials listed of grain size in the μm -, sub- μm or nm range, in which the phenomenon of superplasticity is observed, is claimed to be its major strength. We have adopted this line of argument because then even after conceding that there could be other explanations, which may be able to explain the phenomenon of superplasticity well, one can claim superiority for the present approach because of its universality. As will be seen later,

* Corresponding author. Tel.: +91 40 23134455; fax: +91 40 23012800.

E-mail addresses: kapse@uohyd.ernet.in (K.A. Padmanabhan), Herbert.gleiter@kit.edu (H. Gleiter).

the present approach also will help rationalize what Trelewicz and Schuh [22] have stated on the nanocrystalline–metallic glass transition.

A proper description of boundary sliding pre-supposes a clear understanding of the structure of general high-angle boundaries¹ and the interfaces present in severe plastic deformed materials, metallic glasses, poly-glasses etc., in addition to the boundaries of metallic materials. Our knowledge about the structure of grain boundaries and interfaces in non-metallic materials, metallic glasses and poly-glasses is rather limited. Therefore, the usefulness of the present mechanism for understanding the deformation of the boundaries of the latter classes of materials is due to the rate equation developed being able to fit experimental data pertaining to the different classes of materials equally well, i.e., the justification is based on empirical validation and verification.

2. Structure of high-angle grain boundaries and interfaces

The ideas concerning the structure of high-angle grain boundaries have been derived mostly based on work related to metallic materials. As similar superplastic flow is observed in the other classes of materials also, in the absence of evidence to the contrary, these ideas are considered to be relevant to all the classes of materials that exhibit superplasticity.

Several reviews have examined the models used to describe high-angle grain boundaries for two early reviews, see [23,24]. Of the different approaches, the amorphous boundary model was found to be inadequate,² while the dislocation models were useful for understanding the behavior of small-angle boundaries. Island models were incomplete. The coincidence site lattice model [25–28], which is based on the original suggestion of Kronberg and Wilson [29], considers only geometry. This is surprising because, long before this model was proposed, it was known that all elements of a given structure, e.g., FCC, are not identical and that CSL being purely geometrical, does not even require grain boundaries [23,24]. The structural unit model, in contrast, takes into account geometry and the inter-atomic forces present among the boundary atoms. The latter model suggests that the most stable boundaries are known by minimizing the total free energy of the system [23,24,30–32]. However, there are also situations in which thermodynamic non-equilibrium introduces selection rules where it is not the free energy, but the fast reaction rate, that determines the structure of grain boundaries. Such structures must be related to the maximization of entropy dissipation rate [33,34]. In such cases metastable grain boundary structures will result. It will be seen below that the analysis presented in this paper applies equally well to such boundaries also. Geometrical elegance, often seen in crystalline materials is, therefore, a consequence of thermodynamic/kinetic causes.

Using computer simulations, Gleiter, Chalmers and Weins [23,24,30–32] showed that there were actually no coincidence atoms, but that the parameter that characterized the coincidence boundaries was their periodicity. Even in non-coincidence boundaries, the same atomic groups that characterized the adjacent coincidence boundaries appeared in an alternating sequence. These groups formed two-dimensional arrays of groups of atoms, known as the “structural units”. The atomic structure of any high-angle grain boundary was suggested to comprise a two-dimensional arrangement of these units mixed in a form that corresponds to the orientation relationship of the two crystals that form the

boundary. Needless to say, the accuracy of prediction of the boundary structure in terms of this model will improve significantly if precise inter-atomic potentials are used. While such potentials are available for some elements, which are then used as model materials to test the validity of the theory, the situation is rather bad for engineering materials, often consisting of several alloy elements. Focused research for obtaining precise potentials for the engineering materials will be extremely useful.

The coincidence site lattice (CSL) concept does not also tell one which of the short periodic boundaries is a “special” one, i.e., a low energy boundary. The rotating sphere experiments [35,36] demonstrated that for two materials with identical CSL, the special boundaries could be different. Crystal structure and a consideration of the geometry of grain boundaries alone do not tell one about the thermodynamic stability of a boundary. For example, without knowing the inter-atomic forces, one cannot understand why Ge with diamond structure is brittle, while FCC Ge is ductile or indeed why the same element Ge should crystallize in two different forms. As with crystals, one may compare the properties and structures of boundaries in materials of similar inter-atomic forces. The rotating sphere experiments revealed the same low energy boundaries in Ag, Au, Cu and even Ni.

The approach of combining the CSL concept and the dislocation model of grain boundaries to describe high-angle boundaries [37,38] suggests that when the orientation relationship of the two grains deviates from the ideal coincidence relationship, in analogy with the Read–Shockley dislocation model for small-angle boundaries [39], off-coincidence high-angle boundary is describable as a network of dislocations superimposed on the corresponding ideal coincidence boundary. This idea of describing the “bad fit” regions of the high-angle grain boundary in terms of misfit-dislocations (whose density increases significantly with increasing deviation from ideal coincidence relationship) has been abandoned since 1977 for the following reasons. (a) The suggestion that in some cases the dislocation could dissociate into DSC (displacement-coincidence) dislocations with Burgers vectors too small to be observable using TEM [40] has not been followed up since the 1970s. (b) By postulating that in general high-angle grain boundaries and metallic glasses there is no long range order and so at infinite time the shear modulus would fall off to zero [20], it could be predicted using the Peierls–Nabarro equation that the width of the dislocation (defined by its core) will become infinite (Eqs. (5)–(9) of [41]). This prediction is confirmed by the TEM observation [42,43] that when a lattice dislocation enters a high-angle grain boundary, the dislocation image gradually becomes fainter and disappears. This has been interpreted in terms of splitting of dislocation, spreading and overlap of dislocation cores. (c) It could be shown that solitons, e.g. dislocations, vacancies, get delocalized in high-angle boundaries and metallic glasses [44] and therefore, the presence of dislocations in high-angle boundaries is an unsustainable concept. (d) To the best of our knowledge, MD simulations also have provided no evidence during grain boundary sliding along general high-angle grain boundaries for the presence and glide of grain boundary dislocations. As a result, since 1977 descriptions of high-angle grain boundaries, e.g., [45–59], use only free volume to describe the regions of poor fit.

Thus, in contemporary view, the general high-angle boundary does not have long-range order. It is made of numerous structural units and free volume to give rise to the myriads of general high-angle grain boundaries of different orientations. As already mentioned, the inter-atomic forces present in each case will determine the size, combination and distribution of such short- and intermediate-range ordering of atoms, which is a consequence of the tendency of grain boundaries to minimize the total free energy. The structure becomes less orderly, when one considers metallic glasses and poly-glasses. In these latter cases, the need to preserve

¹ It has been known that grain boundary sliding is the easiest at high-angle boundaries and in fact during the development of superplastic aluminum–lithium alloys, when thermomechanical processing led to microstructures of small-angle grain boundaries only, superplasticity could not be induced (for details, see [20]).

² However, in case of ceramic materials in which strong segregation of glass forming solutes is present, the grain boundary structure could be amorphous.

crystallographic relationships is eliminated, but still short- and intermediate-range ordered clusters of atoms are found, along with free volume. Depending on the degree of amorphousness of the glasses, the density and formation of full or frustrated short-range ordered icosahedral and other types of clusters and clusters of intermediate-order differ and when one reaches a poly-glass the interface between the atom clusters on either side is made mostly of atomic ensembles of short-range order in an otherwise non-crystalline medium. The free volume fraction in this case can be larger than what is found in a general high-angle boundary or a metallic glass [44]. It is emphasized that the arrangement of atoms in the short-range ordered ensembles in all the classes of materials will depend on the inter-atomic forces present among the elements that constitute the material/glass.

In brief, the following ideas are well established so far as interfaces in metallic materials are concerned. In this analysis the same ideas are regarded as valid for the other classes of materials also, so long as there is no evidence to the contrary.

1. For deducing useful information regarding the structure and properties of grain boundaries, it is essential to consider both the boundary geometry and the nature and magnitude of the inter-atomic forces.
2. The CSL model and the method of inserting dislocations into the CSL lattice to account for grain boundary orientation away from the ideal CSL [37,38] are useful concepts for a description of low energy boundaries, including the low-angle variants. But, as the CSL model is based purely on geometry, this method does not suggest a way of knowing the boundaries of low energy. Moreover, it is debatable if under the above restricted conditions the CSL description is any different from the Read–Shockley model for small-angle grain boundaries [39], except that the grain boundary in the latter case is described in the former by the CSL lattice.
3. In case of high-angle boundaries, it is better to discuss in terms of structural units, rather than the CSL.
4. There is experimental and MD simulation evidence for the presence of free volume in high-angle grain boundaries. In all models of high-angle grain boundaries proposed since 1977, the bad fit regions in the boundary are described only in terms of free volume and dislocations are considered to be unstable in such boundaries. It can be shown that free volume and compressibility of a boundary are also material-dependent parameters, i.e., the size and distribution of free volume depend on the inter-atomic forces present among the grain boundary atoms [60].
5. To the best of our knowledge, there is no evidence-theoretical or experimental for the presence of intrinsic/geometrically necessary dislocations and extrinsic grain boundary dislocations in general high-angle boundaries.
6. When a crystal lattice dislocation enters a high-angle grain boundary, that region in the boundary gets more disordered, which cannot be described usefully in terms of ideas used to describe lattice dislocations, e.g., O-lattice, DSC lattice.

3. A mechanism of deformation for disordered states of matter

A dominant view has been that grain boundary sliding is an inherently fast process and that at no stage it can be controlling the rate of deformation [61]. Therefore, attention is focused in many studies on analyzing the process that accommodates grain boundary sliding. Often, based on an experimental determination of activation energy, Q , and the strain-rate sensitivity index of flow, m , in the superplasticity rate equation

$$\dot{\epsilon} = K \sigma^{\left(\frac{1}{m}\right)} \exp\left(-\frac{Q}{kT}\right) \text{ or } \sigma = \dot{\epsilon}^m \exp\left(\frac{Q}{kT}\right) \quad (1)$$

expressed in a dimensionless form, where $\dot{\epsilon}$ is the strain rate, σ the applied stress, k the Boltzmann constant, T the temperature on the absolute scale, and K and K_1 are constants, suggestions are made about the rate-controlling mechanism. In a few cases, microstructural findings are also included in a qualitative discussion. But, in all these cases, as in the analysis of Mukherjee et al. [62], (a) no method of calculating the activation energy for the deformation process and the strain rate sensitivity index using physical concepts has been suggested, and (b) contrary to experimental results, n ($=1/m$), the stress exponent, is assumed to be independent of stress and temperature and Q independent of the value of the (σ/G) ratio, which is maintained constant for determining Q (G is the shear modulus). Moreover, as a result of these studies, different rate-controlling mechanisms have been suggested for different superplastic material systems [2–5]. Consequently, the number of mechanisms and empirical constants needed to explain the phenomenon of superplasticity become very large.

In a contradistinctive approach, it was suggested in 1977 that the rate-controlling process during superplastic deformation in the different types of materials of several grain size ranges is grain boundary sliding (GBS). No doubt, in order to ensure strain coherency, parts of the grain near its boundaries also will be deformed by accommodation processes. The latter process is assumed to be a faster mechanism. (This can now be proved by comparing the rate equations of the accommodation processes with that for GBS.) As there was no consensus on the structure of high-angle grain boundaries in 1977, the original treatment avoided a detailed discussion of the atomistics of the boundary sliding process [8].

All models for high-angle grain boundaries proposed in the period 1977–1984 [45–59] accepted (a) the structural unit concept, and (b) the description of the regions of poor fit in the grain boundaries in terms of free volume. The absence of a need to describe the regions of poor fit in terms of grain boundary dislocations ensures that general high-angle grain boundaries form an interconnected 3D network, which can be viewed as an infinite continuum. In addition, the conclusion of Wolf [58,59] that the excess free volume of high-angle grain boundaries varies only slightly with misorientation allows the development of a deterministic model of grain boundary sliding by ignoring this dependence as a first order approximation. In the early papers, a model was proposed to explain superplasticity in microcrystalline metallic materials [9–13]. But, over time it has been shown to be useful for understanding superplastic flow in the other classes of materials also, regardless of whether the grain size is in the μm -, sub- μm - or the nm-range [9–20,63]. The differences in the analysis for each material and class of materials lie in (a) the frequency and arrangement/distribution of short-range and intermediate-range ordered clusters of atoms and the density and size of the structural units (present in crystalline materials), (b) the size and distribution of free volume in the grain boundaries, and (c) the inter-atomic forces present among the atoms that constitute the grain boundaries. Factor (c) affects significantly the other two microstructural parameters mentioned in (a) and (b). These differences will affect the rate of boundary sliding, the internal stress distribution that will result from boundary sliding and the strain-rate sensitivity of flow (which will depend on the internal stress distribution, as the latter determines the resistance of the material to deformation at any given location). These factors give rise to the differences between the mechanical response of materials of different classes and among materials within a given class. However, in case of poly-glasses, the relative density could be much less than what is found in the other types of materials. This could have a significant effect on the internal stress distribution present in that class of materials

after deformation, which, in turn, may affect the ductility. This point is developed later.

To present this analysis in perspective, it is noted that in a very early view [64] plastic deformation implied deformation over an entire plane – Fig. 1. However, once it was realized that such a deformation mechanism overestimates the theoretical strength of a material by about 1000 times, the concept of dislocation was introduced – Fig. 2. Dislocations are one-dimensional defects introduced to explain 3D flow. In that case, in the plane of the sheet at any instant deformation is confined to distances of 2–3 atomic diameters only. But, in the perpendicular direction, the distance covered during unit deformation can be considerably more. In contrast, deformation of a grain boundary is almost 2D because the grain boundary width is only of 2–3 atomic diameters, while its area often is much larger. To make their analysis consistent with that of Argon [65] for metallic glasses, Padmanabhan and Schlipf [9,10] suggested that the basic unit of sliding will have a radius along the grain boundary plane equal to the grain boundary width, W , and the height, W , of the basic sliding unit will be located symmetrically about the grain boundary plane, with a height ($W/2$) on either side of the boundary. This shape of the basic sliding unit corresponds to an oblate spheroid. There is nothing special about this shape. Argon [65], for example, has chosen the shape to be spherical. We chose the oblate spheroid shape for mathematical convenience, as Eshelby [66] has pointed out that when an oblate spheroid (ellipsoid) gets deformed, it still remains an ellipsoid and by determining the homogeneous stress and strain fields that would develop inside a deformed ellipsoid, it would be possible to deduce useful information on many physical and engineering properties of the material. In contrast, the stress and strain fields outside the deformed basic unit (or inside and outside other deformed shapes) would be highly non-uniform and extremely difficult to determine. In fact, this oblate spheroid shape for inclusion/reinforcement is the preferred kind in the field of composite materials mechanics for determining the average internal stresses [67] resulting from thermal and mechanical effects. Fig. 3 is a schematic of the basic sliding unit that contains free volume.

Based on the deformation of such oblate spheroids how it is possible to calculate the stress, strain, the activation energy for rate-controlling deformation and the stress–strain rate–temperature–grain size interdependence has been explained in a series of publications for many systems [8–20]. As mentioned earlier, the shape and dimensions of the basic sliding unit in the analysis for different classes of materials is the same. However, the atomic

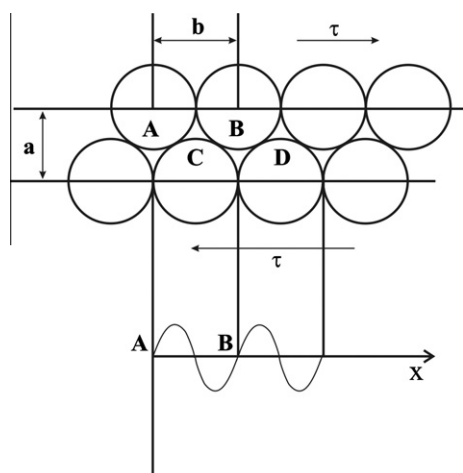


Fig. 1. Frenkel's model for determining the shear yield stress of a material (schematic; from the lecture notes of Strudel).

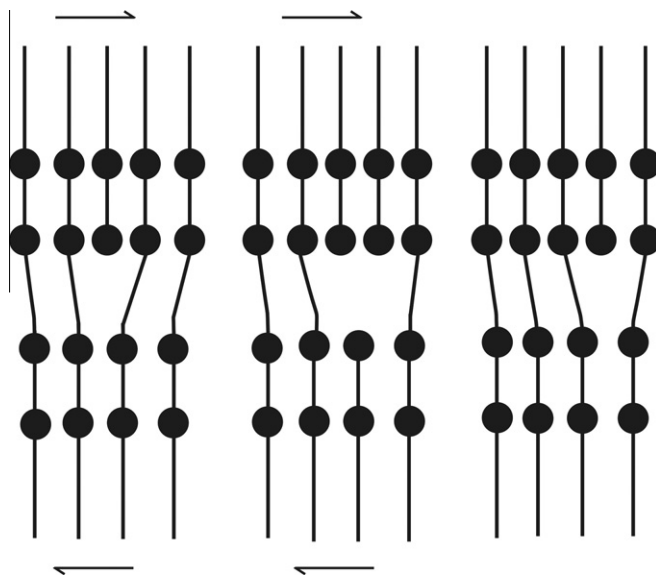


Fig. 2. Disturbed region present during the movement of a lattice dislocation (schematic; from the lecture notes of Strudel).

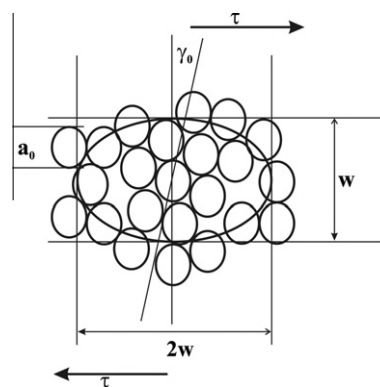


Fig. 3. Sliding/shear unit, according to the present model.

arrangement within this unit and the inter-atomic distance, which will determine the bond strength, the shear modulus, and the free volume will all depend on the nature and magnitude of the inter-atomic forces. With respect to the sequential operation of such unit sliding/shearing events also, which eventually lead to mesoscopic boundary sliding, there are differences, as one moves from a general high-angle boundary of one misorientation to another in a metallic or ceramic material to a composite, a material containing dispersions, a metallic glass or a poly-glass. This is because in the grain boundaries/interfaces of these different classes of materials, short-range and intermediate-range order present in the clusters of atoms, free volume and the frequency and distribution of these microstructural features are dissimilar.

A stress $\tau > \tau_i$ is necessary to achieve unit shear in a basic sliding unit (see Fig. 3 of [9]), where τ_i is the threshold stress needed to initiate an atomic scale sliding event. Wolf's [58,59] finding that excess free volume present in high-angle grain boundaries varies only slightly with misorientation allows this dependence to be ignored as a first order approximation, without introducing serious errors in the calculations. On an experimental basis, this near-misorientation independence of free volume appears to be the case, regardless of the route of manufacture, which could vary from physical/chemical vapor deposition and powder metallurgy at one extreme to severe plastic deformation (SPD) at the other.

Again, based on experiments it also appears to be the case for many bulk metallic glasses in which grain boundaries are replaced by interfaces between atomic clusters (which corresponds to grains in crystalline materials) at which deformation gets concentrated. Careful diffusion studies [68] have revealed that in metallic materials, when substitutional diffusion is involved, diffusivity is the highest at “non-equilibrium” boundaries (produced by SPD), followed by relaxed high-angle grain boundaries and then glassy alloys/metallic glasses, in that decreasing order.³ This observation seems to suggest that the present argument about the commonality among the crystalline metallic and ceramic materials of grain sizes in all ranges and metallic glasses is valid. Such an assumption has two consequences:

1. For stress $\tau > \tau_i$, the atomic configuration found in any basic sliding unit C' is topologically equivalent to C_{eq} (the equilibrium configuration in which the basic sliding unit is found initially) and forms the new equilibrium configuration C'_{eq} . Again there will be metastable states C'' nearby, which are transformed by the action of τ_i to a lower energy state of C'_{eq} equal to C_{eq} , so that grain boundary sliding (GBS) can proceed by a sequence of shear transformations. The equivalence of C_{eq} and C'_{eq} constitutes iso-configurational flow kinetics, which can be described by transition state theory [9,65].
2. The individual shear transformations γ_i and the momentary volume expansions ε_i (as the basic unit goes from one metastable configuration to another, as it is embedded in a solid matrix) will not differ much for different basic units or for the different metastable states within a basic unit.

Using this simplification how the deformation kinetics can be worked out has been explained in detail in the earlier papers [9–20]. Presently it is reiterated that in our view this is the rate-controlling deformation mechanism for superplastic deformation in metallic materials of grain sizes in the μm , sub- μm and nm range, intermetallics, metal-matrix composites and metallic materials containing dispersoids. (The last two give rise to “high strain-rate superplasticity”, i.e., $\dot{\varepsilon} \geq 10^{-2} \text{ s}^{-1}$). As the features of superplastic deformation are very similar in ceramic materials, ceramic-composites, bulk metallic glasses and geological materials also, it is conjectured that a very similar mechanism operates in these materials as well. In poly-glasses, where the free volume fraction could be higher than what is found in metallic glasses, however, the local internal stresses developed by boundary sliding could exceed the fracture strength of the material at certain locations and then there could be premature failure. Another point worthy of note is that dislocation processes can be present dominantly in materials of grain size in the μm range. Therefore, many authors have proposed dislocation mechanisms, rate-controlled by dislocation climb, as the rate-controlling process during superplasticity in materials of μm and sub- μm grain sizes. But, most of these analyses make use of the “largely empirical” equation due to Mukherjee et al. [62] and obtain fits to the data, without presenting *ab initio* physics-based derivations. In addition, for different materials different rate-controlling processes have been suggested. Using Hutter's [21] mathematical proof referred to in Section 1, it is suggested that the present analysis, in view of its universal applicability, could/should be regarded as better than the other approaches,

without even going into the merits and demerits of the other mechanisms. The fact that there are no adjustable constants in the analysis is another point in favor of this approach. It is important to note that the present analysis can apply to all kinds of interfaces that are not amenable to physically realistic crystallographic descriptions and where the fraction of free volume present at the interface is practically independent of the misorientation angle that the coming together of two grains/atom clusters introduces. Based on this deterministic model [8–20], the following key equations have been derived.

Taking the grain boundary width to be ~ 3 atomic diameters, the mean shear strain, γ_0 , resulting from a unit boundary sliding event has been estimated as [9]

$$\gamma_0 \approx \frac{S_0}{2a_0} \text{ or } S_0 \approx W\gamma_0 \quad (2)$$

where S_0 is the local displacement of the atoms in the basic sliding unit in a single sliding event, a_0 the inter-atomic distance and W is the grain boundary width. The mean free energy of activation, ΔF_0 (which replaces a spectrum of free energies of activation at different basic sliding units in the boundary at each of which the free volume fraction could be slightly different), for the rate-controlling process is given by

$$\Delta F_0 = \frac{1}{2}(\beta_1 \gamma_0^2 + \beta_2 \varepsilon_0^2)GV_0 \quad (3)$$

where for an oblate spheroid $\beta_1 = 0.944(1.590 - p)/(1 - p)$, $\beta_2 = 4(1 + p)/[9(1 - p)]$, with p the Poisson ratio, ε_0 the momentary dilatation experienced by the basic sliding unit as it moves from one equilibrium to another equilibrium position under the action of an external shear stress inside a solid matrix, G the shear modulus and V_0 is equal to the volume of the basic sliding unit (an oblate spheroid), i.e., $(2/3)\pi W^3$. ε_0 is related to γ_0 through an appropriate yield criterion, e.g., von Mises, Tresca, Mohr–Coulomb, Gurson, Drucker–Prager etc. When the von Mises criterion is used, $\varepsilon_0 = 0.57\gamma_0$.

The strain rate of deformation, $\dot{\gamma}$, for the rate controlling process is given by [9,13,15–19]

$$\dot{\gamma} = A(\tau - \tau_0) \exp\left(-\frac{\Delta F_0}{kT}\right); \quad A = \left(\frac{2.0944w^4\gamma_0^2\nu}{kTL}\right) \quad (4)$$

where τ is the externally applied shear stress, ν the thermal vibration frequency and L is the average grain size of the material. (In real materials, there would be a grain size distribution. In addition, during superplastic flow, there is grain growth. Therefore, L in Eq. (4) is obtained using standard statistical methods to take care of grain size distribution and a secondary kinetic equation for grain growth.) τ_0 , the threshold stress, which should be overcome to convert sliding at a single grain boundary to mesoscopic boundary sliding (plane interface formation), which is defined to be of the order of a grain diameter or more, is estimated using energy balance. Following Gittus [68], the grain shape is assumed to be rhombic dodecahedron. The surface of the mesoscopic boundary plane will consist of a regular arrangement of triangular pyramids and valleys of height $h = [(6^{0.5}/12)L]$ and ground area $A = [(3^{0.5}/4)L^2]$ (see Fig. 1 of [16]). Leveling of the peaks and valleys by local boundary migration results in an enlargement in grain boundary area per grain, ΔA , perpendicular to the boundary. A decrease in the grain boundary area will be present along the directions of plane interface formation.⁴ These processes need energy expenditure and give rise to the long-range threshold stress, τ_0 . By matching the work done in

³ This trend, however, could change in the presence of interstitial diffusion, segregation or co-segregation to the boundaries, if covalent bonding, instead of metallic bonding, is present among the atoms (about which not much is known), the presence of locally charged layers etc. No systematic study also exists in case of glassy alloys, at least in some of which covalent bonding is important for glass formation (Divinski, personal communication).

⁴ This is a symmetry breaking problem because if the plane interfaces form in the orthogonal direction also, there will be no change in grain shape. (This possibility was brought to the notice of the authors by late Prof. FRN Nabarro.) Such symmetry breaking can result from local differences in grain shape, stress concentration etc.

boundary sliding by the external stress with the energy change associated with the rearrangements in the grain boundary region, the threshold stress needed for plane interface formation/mesoscopic boundary sliding is calculated. In materials of grain size in the μm -, sub- μm and the upper end of the nm-range, where faster (than GBS) dislocation emission from/absorption at grain boundaries will be the dominant accommodation process, τ_0 is given by [13,16]

$$\tau_0 = \left[\frac{8G\Gamma_B r}{3^{0.25}} \right] \left(\frac{1}{L} \right) \quad (5)$$

where Γ_B is the isotropic, specific grain boundary/interface energy and r is the average residual grain boundary/interface misfit removed by diffusion. But, at the very low end of the nm-grain size range in single-phase materials, where (faster) grain boundary diffusion may accommodate boundary sliding

$$\tau_0 = \left[\frac{2^{1.5} G \Gamma_B}{3^{0.75}} \right] \left[\frac{(L - L_0)^{0.5}}{L N^{0.25}} \right]; \quad L_0 = 2 \times 6^{0.5} W \quad (6)$$

where N is the number of grain boundaries that have aligned to form the plane interface. Then, in the range where L_0 can be neglected in comparison with L , Eq. (6) reduces to the much-discussed inverse Hall–Petch relationship [14,15].

4. Some additional predictions

It should be noted that since the latest proposals [17–20] concerning our model, considerable knowledge has been generated on the structure of nanostructured materials, bulk metallic glasses, nano- and poly-glasses. More importantly, these developments have helped us reject the idea that in these materials the atom clusters exhibiting short- and intermediate-range order are packed in random close packing (RCP) (originally suggested by Bernal; see [49] for details) and settle for the view that in these cases also the atomic packing is decided by the nature and magnitude of the inter-atomic forces characteristic of the material. As a result, some more relationships could be predicted and these are summarized here.

Oblate spheroids, in which unit sliding events take place independent of each other, are constructed around free volume sites found in characteristic locations of general high-angle boundaries and the interfaces of metallic, nano- and poly-glasses. Such sites depend on the misorientation of the boundary and the nature and strength of the inter-atomic forces. The unit-sliding event at an oblate spheroid of this kind is depicted in different perspectives in Fig. 4. The direction of shear, as well as the appearance of the ellipsoid before and after the unit-sliding event, is indicated in the figure.

The deformed oblate spheroid, therefore, is the “slipped region”, which is surrounded by the rest of the grain boundary and the matrix, the “unslipped region”. The boundary between these two regions represents a “dislocation”, in the sense in which Volterra defined it. But, in view of what has been discussed already, such a “dislocation” cannot be rationalized in terms of a crystal lattice/structure. It can be seen from the view in elevation of the deformed state that there is a positive closure failure and a negative closure failure respectively of equal magnitude (ΔW) at the left and the right extremities of the deformed ellipsoid, i.e., the deformed state may be considered as having been produced by the passage of a dislocation dipole. Then, the net “Burgers vector” is of magnitude zero if the Burgers circuit is drawn around the deformed oblate spheroid (see p. 13 [69]). In other words, this unit-sliding event may be viewed as a result of the passage of an extrinsic grain boundary dislocation (in the Volterra sense) along the boundary plane, which does not change the boundary misorientation. We have mentioned this merely to point out that our description of the unit sliding event may provide a “justification” for the statement found in the literature that grain boundary sliding takes

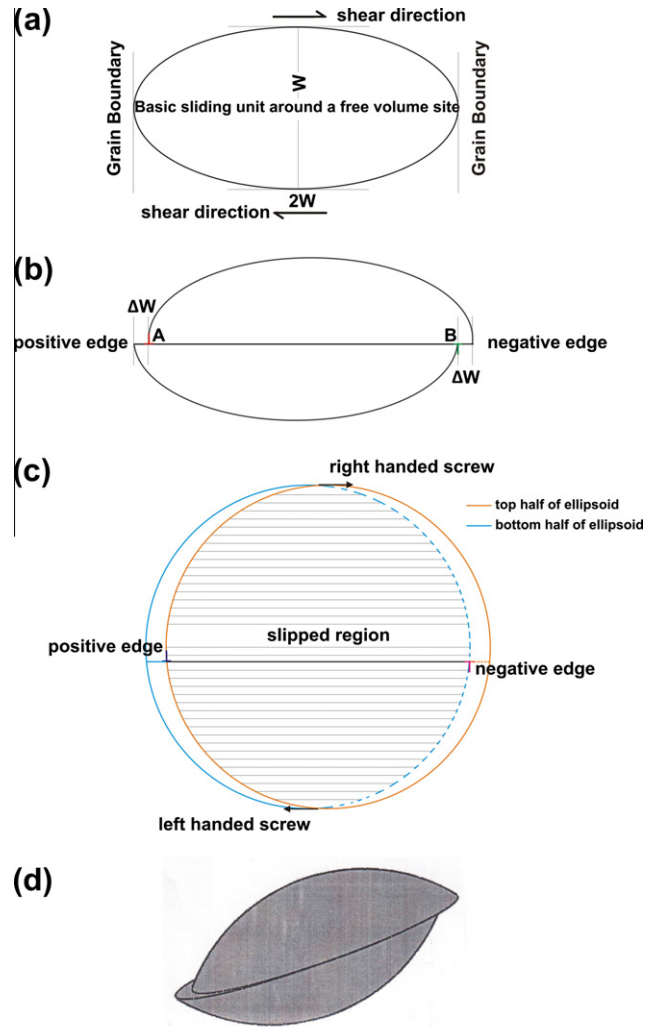


Fig. 4. Description of unit shear event: (a) undeformed oblate spheroid, (b) deformed oblate spheroid, view in elevation, (c) plan view of deformed spheroid, with closure failure at the two extremities indicated in (b) and (c), (d) isometric view.

place by the motion of “extrinsic” grain boundary dislocations along the grain boundary. However, we find the use of this concept as superfluous. We prefer to think of the oblate spheroids as an extremely localized part of a continuum, which, because of the presence of free volume inside it, is of lower density/strength compared with the rest of the grain boundary. Notwithstanding this, it is noted that some authors have found a dislocation-based approach to be useful in deriving some relationships. But, it is not easy to provide a physical justification.

Therefore, the shear strain, γ_0 , due to a single boundary sliding event is given by (see Fig. 4),

$$\gamma_0 \propto \left(\frac{\Delta W}{W} \right) \quad (7)$$

It follows from the theory of elasticity and yielding that for an isotropic solid

$$3\gamma_0 = C \left(\frac{\Delta V}{V_0} \right) \quad (8)$$

where V_0 is the original volume of the ellipsoid ($=2/3\pi W^3$) and ΔV is the free volume present in the basic unit of sliding. $C = \sqrt{3}$ for the von Mises criterion and 2 for the Tresca criterion. Similar

conversion factors (i.e., conversion from the shear to the normal mode of load application or vice versa) can be obtained for the other yield criteria like Mohr–Coulomb, Drucker–Prager, Gurson etc. also. In our approach, flattening of “peaks” or filling of “valleys” (see Fig. 1 of [16]) to form the “plane interfaces”/mesoscopic boundary sliding planes, along which significant boundary sliding can proceed to result in large strains, is achieved by the faster (than GBS) emission of dislocations from grain boundaries and diffusion (details are given in [9,16,70]) which do not enter the rate equation.

From elasticity theory for isotropic solids, it also follows that $\Delta V = 3((1-2p)/E)\sigma_m$, where p is Poisson's ratio, E the Young's modulus of the oblate spheroid and σ_m is the hydrostatic component of the external stress acting on the grain boundary/oblate spheroid.

Hence,

$$\gamma = \left(\frac{C\Delta V}{3V} \right) = \left(1 - \frac{2p}{VE} \right) C\sigma_m \quad (9)$$

As E is much larger than σ_m , the shear strain associated with a unit-sliding event, γ_0 , will be extremely small.

From MD simulations it is possible to calculate E and p , assuming a suitable free volume fraction inside the oblate spheroid and inter-atomic bonding among the boundary atoms appropriate to the material. For uniaxial tension, if the external stress is σ and θ is the angle between the tensile axis and the boundary orientation, then the shear stress, τ , acting along the grain boundary is equal to $(\sigma \cos \theta)$. The hydrostatic component of the stress system is given by $\sigma_m = (\sigma^2 - \tau^2)^{0.5}$. In a uniaxial tension test, this will be tensile in nature and so the hydrostatic pressure will be negative.

In view of the above, ab initio calculation of γ_0 becomes possible. In the numerical validation of the model carried out so far, based on bubble raft experiments and the analysis of Argon [65] for metallic glasses, γ_0 has been assumed to be ~ 0.10 . If the value of γ_0 computed on the lines outlined in this Section is used instead and if that value is different from 0.10, then there would be changes in the computed values of ΔF_0 and σ_0 or τ_0 . The iterative procedure for determining these two unknowns (ΔF_0 and σ_0), which has evolved over time, has been spelt out [13,16–19]. A further progress will be reported shortly. In fact, in the first paper itself [13] it was mentioned that the values of γ_0 (or ε_0) and ΔF_0 have to be determined experimentally (or iteratively, as outlined in the above papers from the experimental stress–strain rate data) and by comparing this ΔF_0 value with the one obtained using the Eshelby equation, the values of γ_0 and ε_0 should be found (see the Appendix of [13])⁵.

Consider any general high-angle boundary (for example, a boundary in a nanocrystalline material, for which MD simulation is possible) that is subjected to a shear stress. In our picture, unit sliding takes place in those regions of this boundary where free volume sites are present. Let the number of sliding sites that actually shear sequentially to cause GBS be N_1 . The magnitude and nature of the inter-atomic forces will determine the magnitude of ΔS , the mean displacement resulting from unit-shear at a basic unit of sliding. Then,

$$\Delta S N_1 \cos \theta = b_1 \quad (10)$$

⁵ In transcendental Eq. (4), there are two unknowns, ΔF_0 and τ_0 . (The equation can also be expressed in the tensile mode by using an appropriate yield criterion. Then, τ_0 will change to σ_0 .) The latest numerical method of solving Eq. (4) is as follows. As $0 \leq \tau_0 < \tau_{\min}$, where $\tau_{\min}$ is the minimum stress at which flow is observed, taking values for τ_0 in the range $0.001\tau_{\min}$ to $0.999\tau_{\min}$ in increments of $0.001\tau_{\min}$, starting from the lowest value, the corresponding ΔF_0 values are found. For each of these τ_0 , ΔF_0 combination, the strain rates are predicted at different applied stresses using Eq. (4) and compared with the corresponding experimental values. The combination of τ_0 , ΔF_0 for which the error in prediction of the strain rate values is the minimum is chosen as the “best” solution. Then, the γ_0 value in the Eshelby equation is chosen such that it is consistent with the “best” estimate of ΔF_0 .

where b_1 is the Burgers vector of the partial (or full) dislocation that will be emitted from a given point in the boundary, θ the angle between the grain boundary at the point of emission of the dislocation and the slip direction of the dislocation into the grain. Thus the point of emission of the dislocation at the grain boundary can be predicted uniquely in terms of Eq. (10). This is a prediction of the analysis that could be verified using MD simulation. Finally, in a real, general high-angle boundary small misorientations between grains could be present locally due to the presence of structural units. In those regions faster dislocation-based grain boundary deformation could be present (as in a small-angle boundary). But, as such regions are bounded by regions of large misorientation containing free volume, the overall deformation rate of the boundary will be governed by the slower strain and strain rate equations derived above.

5. Understanding the deformation of different materials/classes of materials

The manner of analyzing deformation in high-angle grain boundaries of different misorientations in superplastic alloys of grain size in the μm -range (Eqs. (2)–(6)) was spelt out in [9,10,13]. How this analysis could be extended to explain superplastic deformation in nanostructured materials [16], intermetallics [18] and composites and alloys containing dispersoids (high-strain rate superplasticity) [19] was also outlined. In [9,10,13] it was further shown how the analysis could be used to determine the internal stresses that would develop in the vicinity of the grain boundaries in μm -grained materials as a result of grain boundary sliding. That analysis enables one to connect the stress exponent, n ($=1/m$), to the internal stresses that develop due to boundary sliding and thus give a physical meaning to this important practical parameter (m) of superplastic flow. Several studies have revealed that to a good approximation the elongation at fracture increases as the square of the magnitude of the strain-rate sensitivity index, m [2–5]. In this section, the procedures spelt out earlier are generalized to explain how they can be used to connect boundary sliding with m in the different classes of materials listed in Section 1 and in addition how Eqs. (2)–(10) can be used to understand superplastic flow in these materials.

A scrutiny of Eqs. (2)–(10) reveals that ΔF_0 , τ_0 (or σ_0), γ_0 (which is related to ε_0), ΔS , N_1 and b_1 all depend on the composition of the material as well as the nature and magnitude of the forces present among the atoms of the boundary. This is also the case with respect to the two external inputs of G and Γ_B , which are needed to estimate the values of ΔF_0 and τ_0 respectively. Although for mathematical convenience the shape of the basic sliding unit has been assumed to be an oblate spheroid for all classes of materials, the real shape of the basic sliding unit could be irregular and could also change with the composition and the class of material. This can be taken care of by introducing a form factor of the order of unity [9,10]. However, the atomic arrangement within the oblate spheroid and the free volume fraction in the basic sliding unit will depend on the nature and magnitude of the inter-atomic forces. The latter, along with the boundary misorientation, will also decide the distribution and locations of the basic units of sliding and the free volume fraction present along the grain boundary/interface. Thus Eqs. (2)–(10) can be used to understand superplastic flow in any material listed in Section 1. The atomistic details of the present analysis are consistent with the structural unit model [23,24,30–32] and the results of MD simulations [71].

A very important point to note is that not only in the different classes of materials, but also at boundaries of different misorientations in the same material, the internal stresses that develop as a result of boundary sliding can vary widely because of differences

in the arrangement of atoms in the basic units of sliding and the amount and distribution of free volume along the boundary. A method of determining the internal stress distribution at the boundaries of different materials and correlating it to n or m has already been presented [9,13]. Briefly, the procedure is as follows [9,10,13]. Consider any material. The free volume is not distributed continuously along the boundary, but is concentrated at certain locations, which depend on the orientation and misfit of the boundary in these places and the inter-atomic forces present. Therefore, although the magnitude of the unit shear strain will be approximately the same in all cases, the release of the induced shear component of the strain in the vicinity of the basic sliding unit would be different. It is only after the surroundings have been sheared by the same amount that the induced shear will be relaxed completely. The presence of such an internal/“back” stress implies that the free energy of activation for different basic sliding events will not be the same for all the locations independent of time, but will hover around a mean value. Therefore, the sequential shear deformation present in the basic sliding units, which in the model is responsible for mesoscopic boundary sliding, will give rise to an internal stress distribution, $g(\tau_i)$, which is strain-dependent. By assuming the internal stress distribution to be lognormal, [9,13] have expressed the internal stress distribution resulting in a uniaxial tension test as

$$g_1(\sigma'_i) = (\sigma'_i \lambda \pi^{\frac{1}{2}})^{-1} \exp \left\{ - \left[\left(\frac{1}{\lambda} \right) \ln \left(\frac{\sigma'_i}{\sigma_m} \right) \right]^2 \right\} \quad (11)$$

where $g_1(\sigma'_i)$ is the functional form of the internal stress distribution, expressed in the tensile mode. $\sigma'_i = (\sigma - \sigma_i)$, where σ is the external tensile stress that acts on the basic sliding unit “ i ” and σ_i is the threshold stress needed for the occurrence of the unit sliding event in that sliding unit. σ_m is the mean of the stress distribution and λ is the standard deviation. Averaging the internal stress distribution over all the boundaries and connecting the result with the overall stress–strain rate response, it has been shown that [9,10,13]

$$n = \left(\frac{1}{1 - \alpha \left(1 - \frac{\sigma'_0}{\sigma_m} \right)} \right) \left\{ 1 + \frac{u}{\sigma_m} \left[1 + \alpha - \frac{1}{1 - \alpha \left(1 - \frac{\sigma'_0}{\sigma_m} \right)} \right] \right\} \left(\frac{\sigma}{\sigma - \sigma_0} \right) \quad (12)$$

where u is obtained from the relationships $\sigma' = (\sigma_m + u)$, $\sigma' = (\sigma - \sigma_0)$ (σ' is the effective stress acting on the boundary; σ_0 the long range threshold stress needed for the commencement of mesoscopic boundary sliding), $\alpha = (2/\lambda\pi^{1/2})$ and $\sigma'_0 = \sigma_m \exp(-\lambda\pi^{0.5}/2)$, the lower bound of the effective stress range present in the experiments. Therefore, it is clear that once σ_m and λ are known, it is possible to determine the value of the stress exponent, and hence the strain-rate sensitivity index, for any given material in any material class. The value of n ($=1/m$) will give us an idea of the elongation obtainable in the material. In poly-glasses, where the free volume fraction can be more than in other types of materials, because of the internal stress distribution exceeding at certain points along the grain boundary the fracture stress of the given material, premature cracking and failure of the specimen may take place. This will also happen in materials, which under optimal conditions exhibit superplasticity, if the experimental values of grain size, applied stress and temperature are well removed from the desired values and undesirable levels and forms of the internal stress distribution develop. In [13], with the help of data pertaining to three superplastic alloy systems that have grain size in the micrometer-range, the method of determining the σ_m and λ values, the internal stress distribution and the stress–strain rate relation (and, by implication, the stress exponent/strain rate sensitivity index) for a given alloy has been explained. Analyzing the experimental data pertaining to other classes of materials of grain size in any range is straightforward.

5.1. Generalization

This paper deals with the deformation of highly disordered solids. But we see some parallels in case of other properties also and as a result believe that the ideas presented here could be useful in other areas as well. The way we see it, every microstructural feature in crystalline materials has its analog in non-crystalline materials. To illustrate, perfect crystal vis-à-vis the slowly cooled glass; vacancy super-saturated crystal vs. rapidly quenched glass; polycrystalline material and a poly-glass; localized vacancies in crystal and delocalized vacancies in glass; localized inter-crystalline boundaries and localized or delocalized inter-glassy boundaries; the shear bands in glass and dislocation slip on slip planes in crystalline material. The question arises as to the reason for the difference between each of the microstructural features in crystalline and glassy materials. The reason seems to be that solitons exist in crystals, while they are not stable in highly disordered solids. In the latter class of materials the shear modulus approaches zero at infinite time. A consequence is that the disordered solid can deform by local shuffling of atoms (as described in the paper), whereas a crystal can only deform by a lattice compatible process involving the movement of dislocations or vacancies. As a corollary, if a crystalline material is plastically deformed severely, internal stresses are generated, but the pair correlation functions are unchanged. In contrast, when a glass is deformed in a similar manner, the pair correlation functions change, which indicates that after different strains the glassy regions no longer match perfectly between specimens [72, Yu, Shao, Shi, Yu, Hahn, Gleiter, unpublished work].

Taking the comparison forward, we note that the oblate spheroid of our analysis is the analog of the dislocation core where the local atomic motion takes place. The crucial difference in the two cases, however, is that behind the dislocation core all the atoms are in place after the deformation, but behind the oblate spheroid the correlation function is changed after the deformation.

If these arguments based on analogy are correct, the ideas presented in the paper could also be useful in understanding other processes, e.g., diffusion, all processes involving atomic motion in highly disordered solids.

As an example, we cite the case of FeSc nanoglass, in which the atomic distortions in the cores of the defects appear to be much larger than the distortions in the cores of the corresponding defects in a crystalline material. As a consequence, the atomic distortions in the glass/glass boundaries in FeSc nanoglass result in a hyperfine field that is even above the one due to α -Fe. In contrast, the melt-spun glass and the grains and the grain boundaries in a crystalline alloy of the same composition remain paramagnetic, which indicates much smaller distortions [Witte, Fang, Gleiter, Hahn, unpublished work].

6. Concluding remarks

Preoccupation with the “rate-controlling” accommodation mechanism for GBS has prevented many workers from examining the details of the present model carefully. After a highly critical assessment [73,74], more recently it has been accepted [75] that the present mechanism could be important in the nm-range of grain sizes. This is because in this grain size range it has been possible in several studies that used MD simulations to demonstrate the shuffling of atoms, which leads to grain boundary sliding, as predicted by the model. For reasons unknown to the present authors, equal emphasis has not been placed on the TEM (experimental) observation of plane interface formation, which supports the idea of mesoscopic/cooperative boundary sliding leading to superplasticity in nanostructured materials [76–82]. Even on the

fastest computers available today, it is not possible to simulate grain boundaries of length in the micrometer-range. Therefore, it has not been possible to verify the model in the μm -range using MD simulation. But, direct experimental evidence for plane interface formation has been obtained using TEM [11,80] in this range also. Of equal importance in this latter case is the texture randomization result [63], which clearly shows that only grain boundary sliding, accompanied by near-random grain rotation resulting from unbalanced shear stresses at the grain boundaries arising from GBS, can reduce an initial texture of 142 times random to a nearly random texture, without the introduction of any new texture orientations during the texture randomization process (random walk). There will be no grain rotation at all when the boundary shear stresses drop to zero almost instantaneously and diffusion controls the rate of flow [81]. Likewise, slip as a rapid accommodation mechanism has no net effect on the texture present [82]. Texture changes, rate-controlled by dislocation processes, on the other hand, have well understood features, which are quite different from those seen in the superplasticity experiments. A prediction is also made that the angle of grain rotation accompanying superplastic deformation is proportional to the square root of the superplastic strain [63]. The usefulness of the model for understanding superplastic flow in metallic materials of grain size in the μm -range [9,13,16], nanocrystalline materials [16,17,83], intermetallics [18], composites and alloys containing dispersoids that give rise to high strain rate superplasticity [19] has been demonstrated. Recently, it has been possible to modify the deformation mechanism map of nickel by taking the present GBS rate equation also into account, in addition to the mechanisms that have already been considered [84]. By analyzing a few more systems in addition, it is shown that there is a distinct region in the grain size-strain rate-temperature space where GBS is the rate controlling process. These findings will be reported elsewhere.

From Eq. (6) it follows that when the grain size is equal to $2 \times 6^{0.5}W$, the threshold stress necessary for the onset of mesoscopic boundary sliding becomes zero. Then, even with a small perturbation the microstructure could collapse, i.e., there could be a nanocrystalline to the metallic glass transition. If the grain boundary width is taken as equal to 0.50 nm, then this change-over would take place at 2.45 nm. Using different experimental techniques, Divinswie et al. [85] have arrived at a value of 0.50 nm for the width of the grain boundary of a nanocrystalline material. It is interesting (a) that the estimate of 2.45 nm for the nanocrystalline to metallic glass transition in this model is very close to what is suggested by Trelewicz and Schuh [22], and (b) that this value is very close to the smallest grain size that has been produced so far.

In recent times the importance of a coupling between the grain boundary deformation process and boundary migration in an orthogonal direction has been emphasized and it has been suggested that even at high-angle boundaries the Frank–Bilby equation, originally derived for small-angle boundaries, is formally valid [86–88]. A coupling factor, $\beta = (v_l/v_n)$, where v_l is the lateral translation rate along the grain boundary and v_n is the boundary migration velocity in the orthogonal direction, has been defined as a measure of this coupling effect. Careful experimental verification of the model [89–92] has revealed (a) that in the high-angle boundary range, even though the structural dislocations could not be resolved, the Frank–Bilby equation is formally obeyed, which is an effect of the geometry of the problem, (b) that in the high-angle boundary migration rate strongly depends on the misorientation, and (c) a misorientation-dependent, relatively high activation energy is needed for the boundary migration process to take place. (The last two points are consistent with the present model because the boundary misorientation determines the free volume fraction inside a basic sliding unit and hence the free

energy of activation.) There is also one high-angle boundary misorientation ($\theta = 22^\circ$) for which $\beta = 1$ [90].

A survey of literature reveals that this problem of coupling between grain and grain boundary deformation was also addressed by Stevens [93] and Cannon [94] in case of grain boundary sliding – diffusion coupled flow. They concluded that regardless of the rate controlling process, the geometry of the grains defines the value of β . This seems to be the case in the recent works also. In fact this will be the case so long as the overall deformation results from a coupling of grain boundary and grain deformation, both of which in principle could operate independently of each other, except for some accommodating deformation in the less deforming region needed to ensure coherency of strain in the specimen. The presence of an activation energy for the boundary migration process different from that for boundary sliding shows that these two processes are governed by different rate equations and the slower of the two will be rate controlling when they are coupled. (Often, the slower process is the one concerned with grain deformation.)

Evidently, in our model β has a value of unity and minimal grain deformation by faster (non-rate controlling) dislocation emission from the grain boundaries and diffusion in the vicinity of the boundaries take place only to the extent required to ensure coherency of strain. Therefore, there will be no coupling. However, local boundary movement is necessary for plane interface formation (see above) and this will lead to some grain size evolution, but much smaller than in the conventional case notwithstanding the severity of deformation suffered by the superplastic material. This point is reinforced by the recent MD simulation study of Schaefer and Albe [95] of the competition between coupled grain boundary motion and mesoscopic grain boundary sliding in nc-Cu. They found (a) that coupled grain boundary motion contributes to plastic deformation of 3D microstructures and leads to the suppression of grain boundary sliding, and (b) that segregating solutes effectively pin the grain boundaries in place and allow for grain boundary sliding on a mesoscopic scale. In other words, solutes are important to ensure that mesoscopic grain boundary sliding is available as a room/relatively low homologous temperature deformation process in polycrystalline metals. These conclusions are in line with our view that the coupling between grain and grain boundary deformation is a consequence of the geometry of the grains, which could evolve during deformation. The study, in addition, allows us to understand (a) why grain growth (caused by high boundary mobility) is considerable in pure metals, which do not exhibit superplasticity, and (b) why two- and poly-phase microstructures, particularly where the composition of the phases are distinctly different, are most conducive for superplastic deformation. Like in the case of segregating solutes, in these cases also grain boundary mobility is suppressed by the large diffusion distances necessary for the occurrence of grain growth [2–5]. An inference from this discussion is that such a coupling is not likely to be present during “true” superplastic deformation, i.e., where grain growth is minimal/negligible in comparison with the large strains and no rate controlling mechanism, other than grain boundary sliding, is operating (the limiting case; $m = 1.00$).

The present mechanism is proposed for systems that do not have long-range order, which, according to crystallographers, sets crystals apart from glasses and amorphous materials. But, the high-angle boundary (or boundary between atomic clusters) is a feature that is common to both crystalline and non-crystalline (glassy, amorphous) materials. As these boundaries become the low energy, low sigma type, the traditional approach based on dislocation mechanisms becomes important. As the high-angle boundaries deviate clearly away from the coincidence boundary, the present mechanism becomes dominant. In the last section, even a nanocrystalline – glass transition could be suggested. This is in line with the recognition of the condensed-matter physicists

that there is a structural continuum between ideal crystals and amorphous solids, i.e., a variety of ordering can be sampled if one moves along the reaction coordinate of crystallization [96]. Such a view makes it difficult for one to define the different areas of applicability of the present model fully in a straightforward manner. For example, should the deformation of quasicrystals, which have long-range order, but no structural periodicity like the classical crystals (but have quasi-periodicity), be understood as due to the movement of dislocations in hyper space or there also one could look for localized defects, which are too small to be amenable to a dislocation description, and develop a deformation model akin to the present one in three dimensional space? At this stage, we are undecided.

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***Extremely important references to the present paper.*

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